

Blockchain For Large-Scale Internet of Things Data Storage and Protection

Ruinian Li¹, Tianyi Song¹, Bo Mei², Hong Li^{3,*}, Xiuzhen Cheng⁴, Liming Sun³

¹Department of Computer Science, Bowling Green State University, Bowling Green, Ohio, 43403 USA

²Department of Computer Science, Texas Christian University, Fort Worth, TX, USA

³Institute of Information Engineering, Chinese Academy of Sciences, Beijing, China

⁴School of Cyber Security, University of Chinese Academy of Sciences, Beijing, China

Department of Computer Science, The George Washington University, Washington, DC, USA

E-mail: {lir,tsong}@bgsu.edu, b.mei@tcu.edu, cheng@gwu.edu, {lihong,sunliming}@ie.ac.cn

Abstract—With the dramatically increasing deployment of IoT devices, storing and protecting the large volume of IoT data has become a significant issue. Traditional cloud-based IoT structures impose extremely high computation and storage demands on the cloud servers. Meanwhile, the strong dependencies on the centralized servers bring significant trust issues. To mitigate these problems, we propose a distributed data storage scheme employing blockchain and certificateless cryptography. Our scheme eliminates the traditional centralized servers by leveraging the blockchain miners who perform “transaction” verifications and records audit with the help of certificateless cryptography. We present a clear definition of the transactions in a non-cryptocurrency system and illustrate how the transactions are processed. To the best of our knowledge, this is the first work designing a secure and accountable IoT storage system using blockchain. Additionally, we extend our scheme to enable data trading and elaborate how data trading can be efficiently and effectively achieved.

Index Terms—IoT; Blockchain; Certificateless Cryptography

I. INTRODUCTION

Internet of Things (IoT) is an emerging term that describes the ubiquitous connection of everyday objects [1]–[4]. With the dramatically increasing deployment of IoT devices, tremendous interactions among the physical objects are enabled, which brings improved efficiency, accuracy, and economic benefits while reducing human interventions [5]. It is estimated by Gartner that there will be over 20 billion connected IoT devices all over the world by 2020 [6]. The great amount of these devices brings lots of challenges in data storage. How to efficiently store the large-scale IoT data, and how to protect the data are issues of great significance.

IoT applications such as smart grid and implantable medical system, involve tremendous data aggregations. In a traditional cloud-based IoT structure, a centralized cloud server collects and controls all the data, which brings two drawbacks: 1) the cloud server needs very high storage capacity to store the IoT data; 2) sensitive data can be easily leaked from the server. For example, server might trade sensitive data with other entities without notifying the data owner. A decentralized structure will properly handle these issues: Data can be transferred and

controlled in a distributed manner as opposed to that in a centralized structure.

Blockchain offers a convenient platform for distributed data storage and protection. In blockchain, a group of users, also known as miners, work cooperately to create blocks as a public ledger that validate and record transactions. In an IoT application such as implantable medical system, data can be stored in Distributed Hash Tables (DHTs) [7] while the pointer to the DHT storage address can be stored in the blockchain. When an entity requests data from the DHT, the blockchain will decide whether the access can be granted or not, i.e., the authentication of the requester is handled by the distributed blockchain miners instead of a trusted centralized server, which achieves the following advantages:

- 1) *Decentralized Storage*: The IoT data is stored off-chain in a distributed way, and an entity can easily find the storage address through the blockchain.
- 2) *No Centralized Trusted Server*: The access to IoT data is controlled by the majority of the blockchain miners, without any intervention from a trusted server. Users do not need to worry about unauthorized access to his/her data.
- 3) *Traceability and Accountability*: Activities such as accessing and modifying the IoT data, can be recorded by the blockchain. No malicious attempts can be made undetected.

IoT devices have low computational power, and they are not capable of conducting complex computations. *Edge computing* is one way to help mitigate this problem. Edge computing, as opposed to cloud computing, is a method to process data at the network edge, rather than in the remote cloud [8], [9]. Edge computing brings realtime computations and communications by leveraging nearby edge servers. A lot of companies such as Intel, Amazon, and Cisco are developing edge-based services to facilitate IoT development. An edge device could be any computing resource residing between data source and cloud. In our scheme a smartphone or any local computing device can be used as edge server. We assume that the communications between an edge server and IoT devices are secure. This

*Corresponding Author.

is a reasonable assumption as an edge server is placed in the same local network as the IoT devices, aiming to help the IoT devices perform certain kinds of computations. For example, in implantable medical system, an edge server within physical proximity of a patient can collect health data from the implanted sensors. In our scheme, an edge server has two duties: 1) helping the IoT devices perform cryptographic computations; and 2) collecting data from IoT devices and forwarding data to the DHT.

Fig. 1 illustrates the proposed structure for IoT data storage using blockchain. In this structure, a group of IoT sensors send realtime data to an edge device that manages the data storage in DHT through the blockchain. Blockchain works as a “trusted third party” in the following two ways:

- 1) Before an edge device forwards data to DHT, it posts a “transaction” to the blockchain, announcing that the data belonging to certain IoT device will be stored in an address of DHT. Blockchain verifies the transaction and records the identity of the IoT device and the storage address. In this way, blockchain helps manage data storage.
- 2) When an IoT device requests data from DHT, it posts a “transactions” to the blockchain. The blockchain will work as a “trusted third party” to authenticate the requester. If the transaction is validated and written into a block, the DHT node storing the data will send data to the requester. Therefore, authentication is performed through the blockchain, without a trusted server.

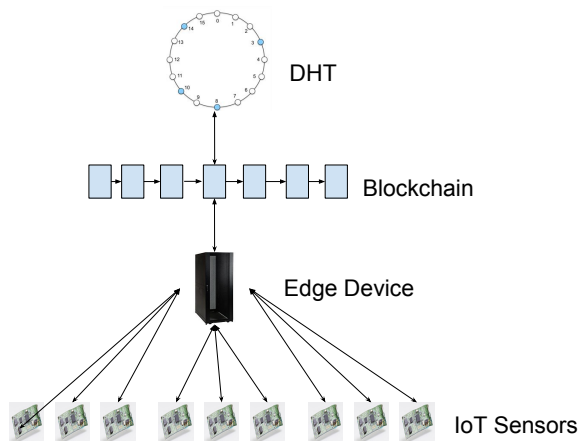


Fig. 1. The structure of data storage scheme with blockchain

Applying blockchain in large-scale IoT data storage and protection is challenging. The most significant issue is how to manage the identities of IoT devices so that authentication can be easily done through blockchain. Note that the miners take charge of authentication when an entity requests to access the data. However, the miners should not have any knowledge of the credentials to perform authentication. This implies that the system must have some cryptographic mechanism that allows an IoT device to be identified and verified by other parties

without utilizing a secret value such as password. Traditional Public Key Infrastructure (PKI) with certificate introduces too much redundancy. Identity Based Encryption (IBE) is one alternative that enables a user’s public key to be created using his identity, so that other entities can verify the user’s identity through the public key. However, IBE suffers from the *key Escrow* problem: the Key Generation Center (KGC) is aware of the user’s private key, and there is no way to authenticate a user unless we assume the KGC is completely trusted. This is where *Certificateless Cryptography* [10]–[12] steps in. Certificateless cryptography is different than IBE as a user’s public key is generated by both the user’s identity and some secret of which the KGC is not aware. Therefore, KGC has no knowledge of the user’s private key, while a public key can be verified whether it belongs to certain user or not. The only drawback of certificateless cryptography compared to IBE is that the public key of a user, even though can be verified, needs to be pre-broadcasted. The good thing is, blockchain offers a platform to share public information, which means that the user’s public key can be shared via the blockchain. For example, an IoT device can append its public key to its data access request and sends the request to the blockchain where the blockchain miners are able to verify the public key of the device.

Our contributions are listed as follows:

- 1) We propose a scheme for large-scale IoT data storage and protection. The scheme eliminates the centralized server and guarantees data protection by letting a large group of blockchain miners control the IoT data.
- 2) We incorporate edge computing in our proposed scheme in order to perform computations for IoT devices and forward the data to DHT for storage.
- 3) We propose to utilize certificateless cryptography in the blockchain-based IoT systems. The public ledger of blockchain system offers a convenient way to broadcast any IoT device’s public key, which overcomes the drawback of a certificateless cryptography system. Meantime, certificateless cryptography highly reduces redundancy brought by traditional PKI and offers an efficient way to authenticate an IoT device. To the best of our knowledge, none of previous research has tackled this problem.

The paper is organized as follows. Section II introduces blockchain, edge computing and the preliminary setting of the system. Section III discusses how authentication are done in the proposed system. In Section IV, we extend our protocol to enable data trading. In Sections V, we study the security issues in the system. In Section VI, deployment issues are discussed. In Section VII, the most related works are summarized. We conclude this paper with a future research discussion in Section VIII.



Fig. 2. Blockchain Network

II. APPLYING BLOCKCHAIN AND EDGE COMPUTING IN IOT APPLICATIONS

A. Blockchain Description

Blockchain works as a peer-to-peer network without a trusted third party. Fig.2 illustrates a typical blockchain network. In this network, a user creates a transaction and sends it to a peer-to-peer (P2P) network. The workers in the P2P network will utilize one agreed algorithm to determine how to write transactions into an empty block, and the transactions will only be validated after they are written into a block. As time goes on, there will be more and more blocks forming a blockchain. Blockchain can be viewed as a continuously growing list of records that can be seen by any user in the network. However, Blockchain is not merely a data structure, but more generally considered as the technology that enables a large group of users agreeing on writing transactions into the blocks. The key point of blockchain technology is how to design an algorithm of consensus, and most designs are based on Byzantine Fault Tolerance (BFT) [13]–[15].

The early version of blockchain, such as Bitcoin, is implemented by a clever mechanism called Proof of Work (PoW). In such a system, each block B_n contains a list T of transactions, a random salt, and the hash of the previous block. A new block can be created only if a random salt is found for a block B_n such that $\text{Hash}(B_{n-1}, T, \text{salt})$ starts with a certain number of 0s. Finding a valid salt is a crypto puzzle that needs lots of computational power, and the computing process is called *Mining*. Once a crypto puzzle is solved, a new block B_n is created and the transactions T can be written into this new block. The smart idea behind PoW is that the transactions in the system are performed and audited by a large group of miners who take great efforts to run this system. Therefore, it is reasonable to assume that the system is secure as long as the majority of miners are honest. However, this scheme does not fit our structure. For example, the most famous cryptocurrency system using PoW – the Bitcoin, is only able to process seven transactions per second [16]. Furthermore, PoW consumes too much computation power without creating true financial welfare, which renders the system too costly for IoT systems.

To tackle the power-consumption issues of the Bitcoin, other mechanisms have been explored to replace PoW such as Proof of Stake (PoS) [17], Proof of Space (PoSpace) [18], and Rem (Intel SGX) [19], etc. We found that Rem is the

most suitable mechanism among them. The idea of “Rem” is to replace the power-consuming “Proof of Work (PoW)” with “Proof of Useful Work (PoUW)” where the miners provide trustworthy reports on CPU cycles devoted to inherently *useful* workloads. PoUW is achieved by adopting Intel’s Software Guard extensions “SGX”, which permits trustworthy code to be executed in an isolated and tamper-free environment, and SGX can prove remotely the result of such executions. To put it simply, this smart idea is to let the miners compute useful work for Intel, and in return Intel provides workers with a proof of their work so that the workers can build a block. Generally speaking, any company can construct similar structures to the Intel “SGX” to outsource its work to the miners. Such a mechanism is practical and is suitable for applications that utilize blockchain.

B. Blockchain Transactions

We are using blockchain as a platform to serve IoT data storage and protection, rather than utilizing blockchain as a cryptocurrency. Therefore, “transactions” in our scheme are different from those in the cryptocurrency schemes such as Bitcoin and Ethereum. A transaction in our scheme is any *request* sent from an IoT device asking for services of data storage or data access. For example, a medical sensor A sends a transaction claiming that it stores data in a certain address $Addr$ in DHT. The transaction can be written in the following form: $T = (ID_A, \text{Timestamp}, \text{Action} = \text{store data in } Addr)$. When a doctor’s implantable medical device (IMD) programmer, which we shall call it B , requests data from A , it will post a transaction to the blockchain, requesting sensor A ’s data stored in DHT in the following form: $T = (ID_B, ID_A, \text{Timestamp}, \text{Action} = \text{access data in } Addr)$. Note that a DHT node in “ $Addr$ ” does not send data to a requester until the DHT node confirms that the transaction of the request has been verified and written into the blockchain.

C. Miner Awards

The proposed system is built upon the blockchain run by a large group of miners. One critical question is how to incentivize miners to work for IoT applications. We believe there is sufficient income for mining the blockchain in the proposed scheme from the following three aspects:

- 1) The proposed structure eliminates the centralized server, and the service fee from a traditional server should be transferred to the miners in the blockchain. This can be done by depositing transaction fees in the blockchain transactions. Once a transaction is done, the miners can get the transaction fee immediately.
- 2) As we discussed before, a practical blockchain mechanism for IoT applications, such as Rem, utilizes miners to compute useful work for large companies, such as Intel. In return, these companies will pay back miners for their work.
- 3) Blockchain itself is a cryptocurrency that creates block awards all the time. Though we are utilizing blockchain for data storage services, the operations of blockchain

will inevitably create block awards that can be split among the miners as their rewards.

D. Edge Computing

An edge device plays a significant role in the proposed blockchain based IoT storage scheme. It not only relays messages/transactions for the IoT devices, but also helps manage data storage and perform computations. We list the roles of an edge device as follows:

- 1) Manage the identities of IoT devices. An edge server stores a copy of identities of all nearby IoT devices and helps each device build a pair of keys for authentication through a KGC.
- 2) Create transactions for IoT devices. A valid transaction should include the signature of an IoT device, and the signing process should be conducted by the edge server. Also, sensitive data should be encrypted before sending out for storage, and edge server can help handle the computations.
- 3) Collect and forward data to DHT. The edge server continuously collects data from nearby devices. It determines the DHT address to store the data and sends the encrypted data to the designated address.

E. Security Model

In our proposed scheme, we assume the communications between an edge device and an IoT device are secure. All cryptographic computations are processed at the edge device to reduce the workload of an IoT device. A KGC is utilized to help establish keys for IoT devices, and this KGC is semi-honest, which means that KGC follows the rule to perform computations, but tries to infer sensitive information collected by the IoT devices. In our design with certificateless cryptography, KGC is not able to obtain any user's private key. Data storage and protection are performed solely by the blockchain, without intervention of any other entity. Therefore, the security of our scheme is based on the security of the blockchain mechanism.

III. AUTHENTICATION OF BLOCKCHAIN TRANSACTIONS

Establishing the asymmetric key pairs using traditional PKI [20] or IBE [21]–[23] both have their limitations, as we discussed before. Efficient authentication for IoT devices should be enforced. With certificateless cryptography, an IoT device can be easily verified. For example, an IoT device posts a transaction signed with its private key SK_A , and appends its public key PK_A and ID ID_A to the transaction. A miner is able to check that: 1) this transaction is indeed signed with a private key associated with PK_A , and 2) PK_A does belong to ID_A . In this way, it can be easily verified that whether the transaction was created by an IoT device with ID_A .

A simpler mechanism adopted in current cryptocurrency also achieves the above result, but it does not fit in our applications. For example, Bitcoin lets each user create a pair of keys (PK_i, SK_i) based on Elliptic-Curve Cryptography (ECC), and a transaction is verified by a signature scheme based

on Elliptic Curve Digital Signature Algorithm (ECDSA). The public key PK_i is hashed twice as an address of the user $Addr$. Therefore, other users are able to check if a public key PK_i belongs to an address by checking if $H(H(PK_i)) = Addr$ [24], [25]. This brings lots of convenience compared to traditional PKI as it eliminates the burden of traditional digital certificates and all users are able to verify a user's public key. However, this system does not achieve user authentication or accountability. Firstly, constantly changing identities will make authentication hard. In a system with full anonymity, the only way to check whether an IoT device has rights to access certain data or not is to verify some credentials only known to this device. However, blockchain is an open ledger and devices' credentials will be exposed. Zero-knowledge Proof [26] is a theoretical solution but hard to deploy. Secondly, a system with full anonymity is not accountable. Therefore, assigning IoT devices with unique identities is necessary to maintain a secure and accountable storage system.

We proposed to utilize certificateless cryptography to achieve both authentication and accountability. In certificateless cryptography, each IoT device's unique identity can be used to create the public key, and all other users are able to verify that the public key belongs to this unique identity. Also, if an IoT device revokes the old key, it is able to create a new public key pair using its unique identity. The new key pair is different from the old one, but it is still bounded with the unique ID.

In the following part, we give a detailed description of certificateless cryptography, and how it can be used in our proposed scheme. To make it simple, we omit the description of an edge device in our algorithms. However, all the computations of an IoT device are performed at the edge device.

A. Certificateless Cryptography

Certificateless cryptography was derived from the IBE in order to solve the key escrow problem. In certificateless cryptography, a key generation center (KGC) creates a partial private key based on a user's identity; the user utilizes the partial private key and its own secret value to establish a private key. Since the secret value is only known to the user, KGC is not able to compute the private key. Therefore, key escrow problem in IBE is avoided. The user also creates the public key based on the secret value and makes it public. In detail, there are five general steps for establishing keys PK_A and SK_A for a user A .

- 1) $Setup(1^\lambda) \rightarrow (K, MSK)$: The setup algorithm takes security parameter λ and returns the system parameters K and a secret master key MSK . This algorithm is run by KGC and only KGC knows the value of MSK .
- 2) $PSKeyGen(K, ID_A, MSK) \rightarrow (PSK_A)$: The partial private key generation algorithm takes the system parameters K , a user A 's identity $ID_A \in \{0, 1\}^*$, and the master key MSK , and outputs a partial private key PSK_A . This algorithm is run by KGC and the output will be transported to entity A .

- 3) $SValGen(K, ID_A) \rightarrow (X_A)$: The secret value generation algorithm takes the system parameters K and user A 's identity ID_A , and outputs secret value X_A . This algorithm is run by the user and X_A will be used to transform the partial private key to a private key. This algorithm is run by the user.
- 4) $SKeyGen(K, PSK_A, X_A) \rightarrow (SK_A)$. The private key generation algorithm takes as input the system parameters K , the partial private key PSK_A and the secret value X_A , and returns the private key SK_A . This algorithm is run the the user and only this user knows his private key.
- 5) $PKeyGen(K, X_A) \rightarrow PK_A$: The public key generation algorithm takes the system parameter K and the secret value X_A to construct the public key PK_A . This algorithm is run by the user and PK_A will be broadcast to the public.

The above steps illustrate how to generate the public key pairs using certificateless cryptography. The function of encrypt, decrypt, sign and verify are described as follows.

- 1) $Encrypt(K, M \in \mathcal{M}, ID_A, PK_A) \rightarrow C \in \mathbb{C} \vee \perp$
This algorithm takes the system parameters K , a message M in finite message space, user's identity and public key, and outputs a ciphertext C in ciphertext space.
- 2) $Decrypt(C \in \mathbb{C}, SK_A) \rightarrow M \in \mathcal{M} \vee \perp$
This algorithm takes a ciphertext C and user's private key SK_A , and outputs a message M .
- 3) $Sign(K, M \in \mathcal{M}, SK_A) \rightarrow Sig$
This algorithm takes system parameters K , a message M and user's private key SK_A , and outputs a signature.
- 4) $Ver(M \in \mathcal{M}, Sig, ID_A, PK_A) \rightarrow Valid \vee Invalid \vee \perp$
This algorithm takes a message M , a signature Sig , user's ID ID_A and user's private key SK_A , and outputs if the signature is valid or not.

There are many ways to achieve certificateless cryptography, with both lightweight encrypting and signing features achieved. The readers can refer to *Al-Riyami*'s scheme as an example for details [10], and we do not give complete constructions of the cryptography scheme here to save space. We also give a definition of the verification function that is used to check whether PK_A belongs to ID_A or not:

$$VerID(ID_A, PK_A, K) \rightarrow Valid \vee Invalid \vee \perp$$

To make the description of our algorithms clear, we generally denote the encryption to ciphertext C of data M with public key PK by $C = E_{PK}(M)$, and decryption of ciphertext C with private key SK by $M = D_{SK}(C)$. Similarly, we denote the signing of a message M by $Sign_{SK}(M)$, and verification of a signature Sig by $Ver_{PK}(Sig)$.

B. How Blockchain Transactions Work in IoT Applications

In this section, we give details how transactions are processed in the proposed scheme. To start, an IoT device needs to register in the blockchain network by contacting KGC for establishing its keys. After successfully obtaining a pair of

keys, an IoT device is able to post transactions that can be verified by the blockchain.

1) *Registration*: To start, KGC broadcasts the system parameters K and keeps a secret master key MSK . All IoT devices should have the knowledge of K . Meantime, an IoT device creates a secret value X_A , and generates its public key using X_A and the system parameter K .

When an IoT device would like to register in a blockchain, it will contact the KGC with its ID ID_A . Upon receiving the request, KGC will generate a partial private key PSK_A for this IoT device, sign ID_A and PSK_A with his private key SK_K , and sends the signed message back to the IoT device. The IoT device will verify if the message comes from the KGC, and if yes, it will generate private key using PSK_A , X_A and K . Note that only this IoT device is able to create the private key because it is the only entity that knows X_A .

Algorithm 1 illustrates an IoT devices' operations in the registration process. $SendRequest()$ and $RecvReq()$ denote the functions of sending and receiving messages, respectively.

Algorithm 1 Device Registration

Input: ID_A
Output: PK_A, SK_A

```

1: procedure KEYGEN( $1^\lambda, ID_A$ )
2:    $X_A \leftarrow SValGen(K, ID_A)$ 
3:    $SendRequest(ID_A)$ 
4:    $RecvReq(PSK_A, Sign_{SK_K}(ID_A))$ 
5:    $V \leftarrow Ver(ID_A, Sign_{SK_K}(ID_A), SK_K)$ 
6:   if  $V = Valid$  then:
7:      $SK_A \leftarrow SKeyGen(K, PSK_A, X_A)$ 
8:      $PK_A \leftarrow PKeyGen(K, X_A)$ 
9:   end if
10:  return
11: end procedure

```

2) *Transactions Description and Verification*: After successfully registered at the blockchain, an IoT device is able to take advantage of the blockchain to store data. Generally, a transaction includes the identity of an IoT device, a timestamp and an action, for example, $T_A = (ID_A, Timestamp, Action)$. An action can be a claim to store data at a DHT address, a request to access data, or a request to update data. To save space, we use $Addr$ to represent actions on the DHT address. To verify a transaction T_A , the miners have to check the following requirements:

- 1) If the the public key PK_A is derived from the identity ID_A associated with it.
- 2) If the signed transaction can be verified with the public key PK_A .

By checking the two requirements above, the miners are able to verify whether a transaction T_A is created from ID_A or not. Algorithm 2 illustrates how to verify a transaction with a procedure $VerTrans()$. When a transaction is posted, it is automatically marked with a flag $s = 0$. The flag of successfully verified transactions will be set to "1", and then the verified transactions will be written into a block.

Algorithm 2 Verify A Transaction

Input: T_A, σ_{T_A}

Output: a verified T_A

```

1: procedure VERTRANS(  $T_A, \sigma_{T_A}$  )
2:    $s \leftarrow 0$ 
3:    $V_1 \leftarrow VerID(ID_A, PK_A, K)$ 
4:   if  $V_1 = Valid$  then
5:      $V_2 \leftarrow Ver(T_A, \sigma_{T_A}, ID_A, PK_A)$ 
6:   elseAbort
7:     if  $V_2 = Valid$  then
8:        $s \leftarrow 1$ 
9:     elseAbort
10:    end if
11:  end if
12:  return
13: end procedure

```

3) *IoT Data Storage and Protection:* We take implantable medical applications as an example. The proposed scheme works for various IoT applications such as smart grid, smart home etc. For example, in smart grid, smart meters send data via edge devices to blockchain storage and server/aggregator can access data on billing date. In smart home, personal data generated by IoT devices can be sent to the blockchain storage and the data owner/authorized user is able to access data.

Firstly, a medical device creates an access control list through the edge server, which specifies clearly who can access his data. Then it creates a transaction $T_A = (PK_A, ID_A, ACL, Addr)$ with its public key, identity, access control list and storage address, creates a validation σ_A appended to the transaction, and publishes T_A to the blockchain system. The process is illustrate in Algorithm 3.

Algorithm 3 Store Data

Input: ID_A, ACL

Output: a verified T_A

```

1: procedure SETACL( $ACL$ )
2:   Create  $T_A = (PK_A, ID_A, ACL, Addr)$ 
3:   Broadcast ( $T_A, \sigma_A$ )
4:   return
5: end procedure
6: procedure VERTRANS( $T_A, \sigma_A$ )  $\triangleright$  run by the miners
7:    $s \leftarrow 0$ 
8:    $V_1 \leftarrow VerID(ID_A, PK_A, K)$ 
9:   if  $V_1 = Valid$  then
10:     $V_2 \leftarrow Ver(T_A, \sigma_A, ID_A, PK_A)$ 
11:  elseAbort
12:    if  $V_2 = Valid$  then
13:       $s \leftarrow 1$ 
14:    elseAbort
15:  end if
16:  return
17: end procedure

```

The miners in the blockchain system, once received the transaction T_A , will have to verify the validation of the transaction by checking if the signature is valid, and if the message is signed with the transaction creator's public key. If the verification passes, the flag on T_A will be set to "1," and T_A will be written into new block. If a doctor's device with ID_B would like to access the data, it can create a transaction $T_B = (ID_B, ID_A || Addr)$, signs T_B with its private key SK_B and appends the signature to T_B . The miners in the system will verify firstly if T_B is validated, and secondly if the identity of ID_B belongs to the access control list. Both the two verification need to be passed in order to validate this transaction. The process is illustrated in Algorithm 4.

Algorithm 4 Access Data

Input: $ID_A || Addr, ID_B$

Output: a verified T_B

```

1: procedure REQUESTDATA( $ID_B, ID_A || Addr$ )
2:   Create  $T_B = (ID_B, ID_A || Addr)$ 
3:    $\sigma_{T_B} \leftarrow Sign(K, T_B, Sk_B)$ 
4:   Broadcast ( $T_B, \sigma_{T_B}$ )
5:   return
6: end procedure
7: procedure VERTRANS( $T_B, \sigma_{T_B}$ )  $\triangleright$  run by the miners
8:    $s \leftarrow 0$ 
9:    $V_1 \leftarrow VerID(ID_B, PK_B, K)$ 
10:  if  $V_1 = Valid$  then
11:     $V_2 \leftarrow Ver(T_B, \sigma_{T_B}, ID_B, PK_B)$ 
12:  elseAbort
13:    if  $V_2 = Valid$  then
14:      if  $ID_B \in ACL$  then
15:         $s \leftarrow 1$ 
16:      elseAbort
17:    end if
18:  end if
19:  return
20: end procedure

```

In the proposed scheme, only transactions passing access control can be written into a block. A DHT node that stores data will check if the transaction requesting data exists in the blockchain before it can send data to the requester. If an unauthorized IoT device tries to access sensitive data of a patient, it will be blocked by the blockchain as it can not pass the verification. Note that the access control list is determined by the data owner himself and nobody is able to modify it. Therefore, no entity is able to access data without the data owner's permission. Furthermore, the security of such a scheme is based on the majority of miners, which guarantees the protection of sensitive medical data.

IV. EXTENSION TO DATA TRADING

With the proposed scheme, any user can easily trade his data through the blockchain. Data trading can be done in a transparent and accountable way. For example, a user would

like to sell his IoT data collected by electrocardiogram (ECG) sensors for years to some research institutes. This user can post a transaction through the network edge to the blockchain claiming to sell his data. An interested party can post a transaction, in which it commits a deposit to request the data. If the data owner thinks the deposit is enough, he can post a transaction with an updated access control that includes the requester. The blockchain will verify if the requester can access the data, and if yes, the deposit will be sent to the data owner.

Fig 3 illustrates how data trading is processed. Firstly, data buyer posts a transaction $Deposit_A$ with a valid signature $Sig_{sk_A}(Deposit_A)$ appended to it. The transaction includes an input of the buyer's identity ID_A , the public key of ID_A , and the seller's identity ID_B . The buyer will make a commitment of d dollars, which is locked until certain situations are satisfied: 1) the data seller publishes a transaction T_B that writes the buyer's identity into the access control list of requested data, then the d dollars will be sent to the seller; or 2) after t time, the seller does not sell the data, then the buyer can get his deposit back.

If the seller would like to sell his data to the buyer, he can post a transaction T_B through the edge device to update his access control list by including the buyer's identity inside the access control list for a period of t time. Then the seller is able to post a transaction $Getpaid_B$ that unlocks the d dollars in $Deposit_A$. Note that only the buyer is able to unlock the d because the blockchain will strictly check the following two requirements: 1) if T_B specifies that $ID_A \in ACL_B$ for t time; and 2) if $Getpaid_B$ is indeed created by the device ID_B . If the seller does not sell his data within t time, buyer is able to redeem his money by posting a transaction $GetDeposit_A$. The blockchain will verify 1) if it is the buyer himself redeeming the money; and 2) if t time has passed since the post of $Deposit_A$. If the verification succeeds, then $GetDeposit_A$ will be validated and written into blocks, such that the buyer can get his deposit back.

In the above example, data trading is achieved in an effective and efficient way. The trading process is controlled and audited by the blockchain, which makes trading transparent and accountable. Furthermore, through this mechanism, a seller can quickly obtain trading fee from the buyer by redeeming the deposit transaction, without waiting for a period of time.

V. SECURITY

A. Protocol Security

Theorem 1: Algorithms 1 to 4 form a secure protocol in authentication under the assumption that the adopted certificateless cryptography algorithm is secure.

Proof The security of authentication in the proposed scheme is based on the certificateless cryptography. There exists lots of schemes that are $IND\text{-}CCA$ secure. In such a scheme, an adversary has a negligible "advantage" to distinguish two distinct plaintext from a ciphertext with probability $1/2 + \epsilon(k)$, where k is a negligible function in the security parameter k .

That is to say, an adversary is not able to guess the private key of a user even from large number of ciphertext in the IoT system. Therefore, it is impossible for an adversary to forge a digital signature $\sigma(T)$ for any transactions T published in the system. Furthermore, to forge a public key PK_A of a user ID_A is also not possible, as the fake key cannot pass the verification function $VerID$.

B. Privacy

To protect sensitive IoT data, necessary encryption needs to be carried. Data sent out from an edge device to the final storage can be sent in an encrypted form such that eavesdroppers in the network cannot get sensitive data. The data can be encrypted under the IoT device's public key or under a specified entity's public key. When sharing data to another entity, the data owner could choose to encrypt the data under the entity's public key, or he could choose to use re-encryption [27]. Re-encryption is a useful cryptography primitive that enables data encrypted under one public key to be transformed to data under another public key, without decrypting the message. Data sharing will become easy with the incorporation of re-encryption.

If the IoT data is encrypted when stored, data trading can become more complicated than the example we gave, as the buyer has to make sure that the requested data has been transformed to ciphertext under his own key before making payments. Therefore, re-encryption has to be performed by the DHT node that holds that data, instead of the data owner himself. Particularly, one more step should be added: when the data owner posts a transaction to sell the data, the DHT node re-encrypts the data and posts a transaction to the blockchain claiming that the data is now encrypted under the buyer device's public key. Only after this step, data owner is able to get payment from the buyer's deposit.

C. Traceability and Accountability

The proposed scheme brings traceability and accountability. Any IoT device accessing data in a certain DHT address will be recorded and there is no way to deny this operation. Data owner will know which entity accessed his data, and he is able to make sure none unauthorized entity has accessed his data. Also, malicious attempts to access data will be recorded and detected, which could largely mitigate Denial of Service attacks. When a malicious device constantly challenges the system to access the data, this device can be easily detected and recorded, and will be blocked permanently.

D. Blockchain Security

It is clear that the security of the proposed scheme is based on the security of blockchain and certificateless cryptography. The security of blockchain lies on the hardness of preventing sibyl attacks. In a blockchain, if an adversary is powerful enough to take over the majority of the nodes, then it can perform arbitrary malicious operations on the transactions. Therefore, whether a blockchain system is secure enough depends on if we can incentivize sufficient large number of

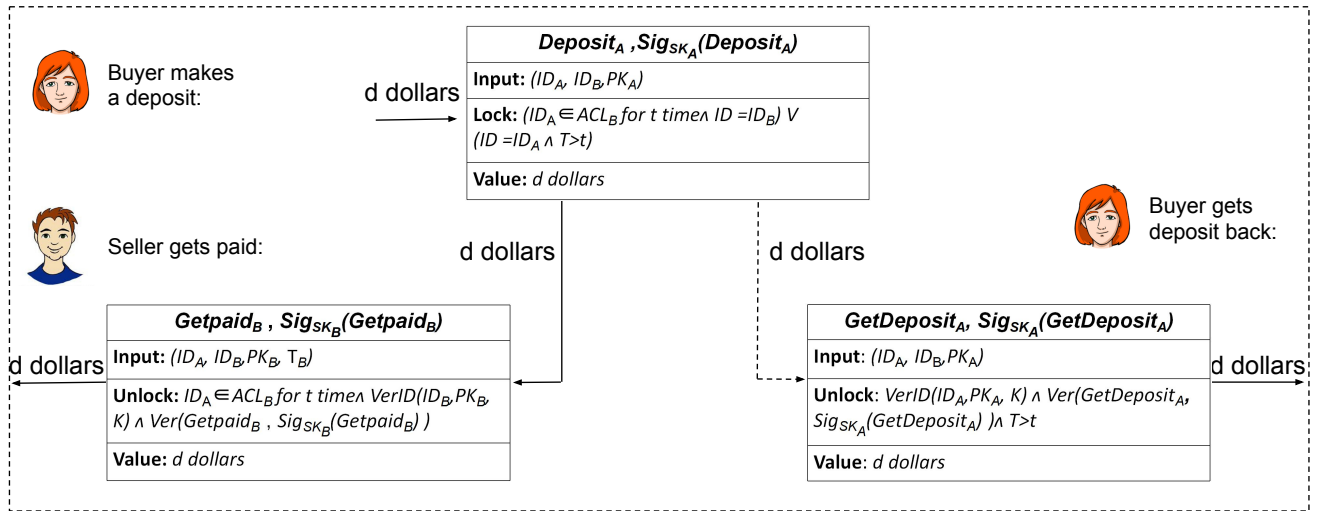


Fig. 3. Data trading with the blockchain

miners in the blockchain. As discussed before, our system brings mining award from transaction fees of data storage and protection, apart from block award in a traditional blockchain cryptocurrency system. Therefore, miners will be attracted in building such a storage system for IoT applications. Furthermore, a single blockchain can work for various IoT applications. The more IoT applications adopt blockchain, the more transaction fee there will be, and the more miners will be attracted to the blockchain. Therefore, the blockchain platform serving IoT data storage and protection will be secure.

VI. DISCUSSIONS

The deployment of the proposed system relies heavily on the design of blockchain system. Without highly scalable blockchain system, it is hard to deploy the IoT storage system. This is the trade-off brought by eliminating the centralized server. Scalability is a major problem in blockchain's design. There are a lot of ongoing research working on it [14], [15], [28]–[30], and two most studied mechanisms to solve the scalability problem are Sharding [31], [32] and sidechains [33]. To find the most scalable system that exists currently, we have to evaluate the blockchain systems that have already been deployed in very large networks. Some very new blockchain technologies such as Ripple [34] could process thousands of transactions per second, but the design is partially decentralized. Decentralized designs such as Ethereum [35] could take only thirteen transactions per second. However, there exists new-emerging blockchain design built based on Ethereum that have much better scalability. For example, Tomochain is supposed to process up to one thousand transactions per second [36]. Supposing a blockchain mechanism is adopted that could process around 1000 transactions per second, then 10,000 transactions could be processed in roughly 10 seconds in the IoT storage system. One good thing is that, with the development of blockchain technology, it can be foreseen that

more and more blockchain designs suitable for the proposed IoT storage system will come out.

VII. RELATED WORK

The rapid growth of Internet of Things promotes the sharp growth of data, and brings lots of challenges in big data storage, analytic and management [37]. For data storage, most researchers focus on building databased management models to mitigate the massive IoT data storage problem. NoSQL [38], [39] and Hadoop [40] databases are attracting most of the attention. Besides, the authors in [41] proposed a new IoT data storage platform based on combined multiple database models. Our work is distinct from these researches as we focus on constructing a novel distributed data storage system based on blockchain technology. To process the massive IoT data, edge computing has become a popular enabled technology [42]. Edge computing and IoT have been extensively studied [8], [43], [44]. The visions and challenges in edge computing are comprehensively studied in [45], and the security and privacy issues in edge computing are extensively discussed in [46].

Blockchain, as a decentralized cryptocurrency mechanism, has received lots of attention. Most research are focused on the blockchain mechanism themselves, aiming to solve the scalability problem [14], [15], [29] or reducing the power consumption of a PoW [17], [19]. There are some research utilizing blockchain on practical applications. In 2014, for the first time, Bitcoin is proposed to form a lottery protocol as a multiparty computation method [47]. The authors demonstrated that Bitcoin provides an attractive way to construct a “timed commitments” and a multiparty protocol can be constructed by letting the miners emulating a trusted server. Bitcoin was also proposed as an incentive mechanism for distributed P2P application [48], in which the relay of the network helps to transfer messages and gets reward from the Bitcoin system. In 2016, Christidis *et al.* discussed how blockchain could possibly work in the IoT domain. They

illustrated several IoT applications that blockchain might fit in, but did not give detailed solutions [49]. The most related work is [50], in which Zyskind *et al.* proposed a decentralized personal data management system using blockchain and illustrated how blockchains could become a vital resource in trusted computing [50]. This work assumed the existence of a secure channel through which the data requester shared secret keys with the data owner. The authentication was processed through the blockchain, and only the users holding the secret keys are able to access data. The work has its limitations as 1) the assumption of such a secure channel is not practical; 2) the secret keys will be leaked to the blockchain miners who perform the authentication. In our research, we take advantage of certificateless cryptography to achieve effective and efficient authentications without a secure channel.

Certificateless cryptography was proposed in 2003 [10] to overcome the key escrow problem in identity based encryption. Compared to traditional PKI, certificateless cryptography does not require certificates to guarantee the authenticity of public keys. Followed by this pioneer research, Seo *et al.* proposed a more efficient way to construct certificateless cryptography without using pairing [51]. The most Efficient Certificateless signature scheme to date was proposed that involve only simple hash operations for signing [52]. Also, certificateless signcryption algorithm was proposed in order to reduce the computation cost and encryption and signing, such that it can be easily carried by the lightweight IoT devices [53]. The above proposed protocols suffer from one problem: it is hard to broadcast the public key of a user. Our proposed scheme with blockchain overcomes this problem as blockchain is a perfect media for spreading public keys to all users in one network.

VIII. CONCLUSION

In this paper, we propose a secure scheme for IoT data storage and protection based on blockchain. Edge computing is incorporated to help manage data storage and small IoT devices perform computations. Certificateless cryptography is adopted to set up a convenient authentication system for the blockchain-based IoT applications, and blockchain overcomes the drawback of certificateless cryptography by offering a platform for broadcasting the public key of a user. We give detailed algorithms on how to process transactions in such a system and how to achieve authentication and accountability. To the best of our knowledge, this is the first paper tackling the problem of building a secure and accountable storage system for large-scale IoT data, and the first to combine edge computing, certificateless cryptography, and blockchain as a whole to serve IoT applications.

Our future work lies on improving our authentication scheme for blockchain based system. In our current scheme, authentication is done by verifying the identity of the data requester. For a system with complicated access control policies, more comprehensive designs need to be explored.

ACKNOWLEDGMENT

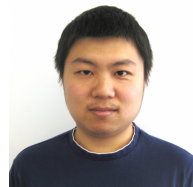
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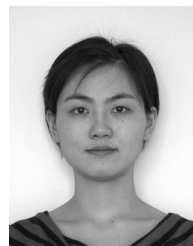
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Ruinian Li received her Ph.D. degree in computer science from the George Washington University in 2018. He is currently an assistant professor in the department of computer science in the Bowling Green State University. His research interests include Internet of Things (IoT), network security, applied cryptography, privacy-aware computing, and blockchain technology.



Tianyi Song received her Ph.D. degree in computer science from the George Washington University in 2017. She is a full-time faculty in the department of computer science in the Bowling Green State University. Her current research interests include secure and privacy-aware computing for the Internet of Things (IoT) and Cyber Physical Systems (CPS), mobile computing, wireless networking, and blockchain technology.

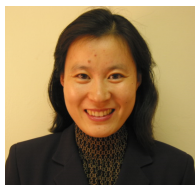


Bo Mei received her Ph.D. degree in computer science from the George Washington University in 2018. He is currently an assistant professor in the department of computer science in the Texas Christian University. He began his research focusing on Mobile Computing since 2014, and has conducted extensive study on system applications for IoT devices. Currently, his research spans the broad area of mobile computing, IoT, and social network inference attacks.





Hong Li obtained his Ph.D. degree in 2016, from the University of the Chinese Academy of Sciences. He received his B.A. from Xian Jiaotong University. He currently works under Prof. Limin Sun in the Institute of Information Engineering, Chinese Academy of Sciences. His primary research interests include security and privacy in wireless networks, and localization.



Xiuzhen Cheng received her M.S. and Ph.D. degrees in computer science from the University of Minnesota Twin Cities in 2000 and 2002, respectively. She is a professor in the Department of Computer Science, The George Washington University, Washington, DC. Her current research interests include privacy-aware computing, wireless and mobile security, cyber physical systems, mobile computing, and algorithm design and analysis. She has served on the editorial boards of several technical journals and the technical program committees of various

professional conferences/workshops. She also has chaired several international conferences. She worked as a program director for the US National Science Foundation (NSF) from April to October in 2006 (full time), and from April 2008 to May 2010 (part time). She received the NSF CAREER Award in 2004. She is a member of ACM, and a Fellow of IEEE.



Liming Sun received the MS and PhD degrees from the National University of Defense Technology, Changsha, China, in 1995 and 1998, respectively. From June 1998 to June 2000, he was a postdoctoral fellow in the Institute of Software, Chinese Academy of Sciences, Beijing, China. He is currently a professor at Institute of Information Engineering at Chinese Academy of Sciences, and School of Cyber Security at University of Chinese Academy of Sciences. His research interests include wireless sensor networks, vehicular ad hoc networks,

and wireless broadband access networks.